

A Fast Calibration of Laser Vision Robotic Welding Systems Using Automatic Path Planning

Peiwen Yang^{ID}, Ting Lei^{ID}, Chaoqun Wu^{ID}, Song Zhao^{ID}, and Jiahao Hu^{ID}

Abstract—Calibration is a crucial issue in laser vision robotic welding systems (LVRWSs). Traditional calibration methods rely on complex manual teaching and fixed pattern of robot end-effector poses, which are time-consuming and will occasionally fall into local optimal solution. To address these issues, a fast calibration method based on automatic path planning is proposed in this article. By ensuring that the camera's optical axis coincides with the center of the calibration board regardless of the robot end-effector's attitude, images covering all feature points will be collected. Based on the camera optical axis constraint, we establish the robot end-effector motion constraint equation. Then, an automatic path planning method based on optical axis constraint equation, rotation constraint, and motion selection strategy is proposed, which improves the diversity of the calibration data. During data acquisition, a hand-eye parameters optimization process is performed simultaneously by minimizing the reprojection error of visual feature points. The actual calibration tests show that the proposed method can significantly improve the calibration efficiency and robustness. In addition, a variety of weld measurement experiments, such as S-type and C-type welds and fillet weld, were conducted under arbitrary welding torch's attitude to evaluate the accuracy and robustness of the method. The mean absolute error (MAE) and root mean square error (RMSE) of the position are within 0.2 mm.

Index Terms—Calibration automation, hand-eye calibration, laser vision robotic welding system (LVRWS), path planning.

I. INTRODUCTION

LASER vision sensing technology has been extensively applied in robotic welding [1], [2], [3], [4]. The technology shows good robustness under strong noise and can obtain the information of welding workpiece such as start point [5], characteristic parameters [6], [7], and joint type [8].

To improve deployment efficiency and reliability, the demand for fast calibration in the intelligent robotic manufacturing is increasing rapidly. The slight change of external parameters is due to the mechanical vibration or deformation of the equipment during long-term operation, which has a direct impact on the accuracy of the robot welding system.

However, traditional calibration methods of robotic welding system rely on tedious manual teaching. Therefore, a teaching-free method is urgently needed, and path planning is a key challenge for calibration automation.

For laser vision robotic welding system (LVRWS), some calibration methods have been presented. Wang et al. [9] proposed an efficient calibration method based on on-site operation toward six robot poses, which required a robot teaching process. In addition, Antonello et al. [10] designed a two-stage hand-eye calibration procedure, which moved the camera to several poses. When the posture of the calibration board changes, redesigning the teaching point is inevitable. In [11], based on deep reinforcement learning, a point-to-wrist network was presented which established an accurate conversion relationship from the pixel frame to the wrist frame of the robot. Zou et al. [12] used calibration data to optimize the calibration parameters with the designed generative adversarial network (GAN). The above data-driven methods rely on massive data by scanning the workpiece in a wide variety of poses. Xu et al. [13] presented a two-stage least squares approach based on detecting straight edges of common objects. Points extracted from the same edge, under various robot poses, were collected to solve the calibration problem. In the method, it was ineluctable to design the robot path according to the placement conditions of edges. Another part of the current research used a special calibration object to construct constraint equations based on point cloud registration algorithm [14], [15]. In [16], the calibration methodology used a circular array of scan lines of laser scanner which were then used to define a set of associated robot poses during the calibration. However, when the hand-eye parameters are unknown, it is impossible to calculate the corresponding poses of the robot according to the laser stripes pattern. To sum up, data collection is a key link to improve the efficiency of hand-eye calibration. However, it relies on the prior information of hand-eye parameters or field environment, and it requires calibrator's experience.

The solving of hand-eye parameters has been studied extensively. Wu et al. [17] proposed a unit-octonion representation through which an optimal closed-form eigen-decomposition solution is derived. Horn et al. [18] solved hand-eye problem based on dual quaternions and provided a certifiable globally optimal solution. In [19] and [20], hand-eye parameters were estimated via optimization on the special Euclidean group SE(3). All the methods above can be used to obtain high accuracy calibration, but they are not suitable for real-time

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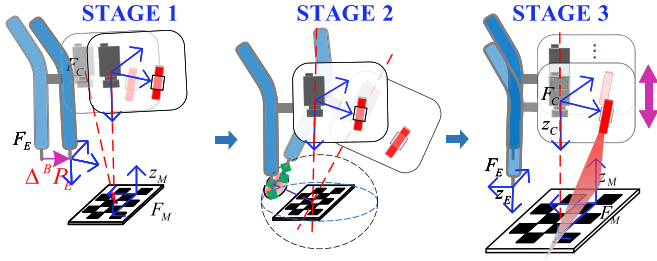


Fig. 1. Calibration process of LVRWS is divided into three stages: 1) initial position adjustment; 2) automatic hand-eye calibration; and 3) automatic line laser plane calibration.

optimization. Moreover, these methods need to obtain the pose sequence of the calibration board first. Yang and Shen [21] applied a maximum posterior estimator to achieve the extrinsic calibration between the inertial measurement unit (IMU) and camera. Koide and Menegatti [22] presented a graph model based on the reprojection error minimization for hand-eye calibration. The method of simultaneous measurement and optimization is more efficient comparing with postprocessing methods.

Motion selection is essential to improve the observability of hand-eye parameters, because degenerate motions or small rotations will introduce significant errors in hand-eye calibration. To eliminate the influence of degenerate motion, Zhang et al. [23] established a degenerate model of hand-eye system, which derived a degenerate form of the hand-eye parameters. Nevertheless, it is hard to expand to common six degrees of freedom (DoFs) robots. An algorithm of motion selection for hand-eye calibration was adopted to avoid the degenerate cases [24], [25]. The “good rules” for motion selection provide a theoretical foundation for our path planning algorithm.

Besides hand-eye calibration, the calibration of self-made LVRWS also includes camera calibration [26] and line laser plane calibration [9], [27], [28], [29]. Obviously, the robot can be used as the motion platform of the laser vision sensor to collect calibration board images and laser stripe images.

When the hand-eye parameters are unknown, mathematical formulas must be established to constrain the spatial relationship between the robot end-effector and the calibration board. Therefore, in this algorithm, we propose an optical axis constraint to solve the problem of collecting effective data, because the traditional algorithms generally requires prior knowledge of the on-site environment or hand-eye relationship. Meanwhile, to generate information-rich robot paths, motion selection strategies must be considered.

This study proposes a fast calibration method based on automatic path planning for line laser vision sensors in robotic welding systems.

The contributions of our study are as follows.

- 1) An automatic calibration scheme for LVRWS is presented in Fig. 1. This method can automatically plan the end-effector’s path and optimize system parameters of LVWRS while collecting data.
- 2) By aligning the optical axis of the camera with the center of the calibration board, the motion constraint

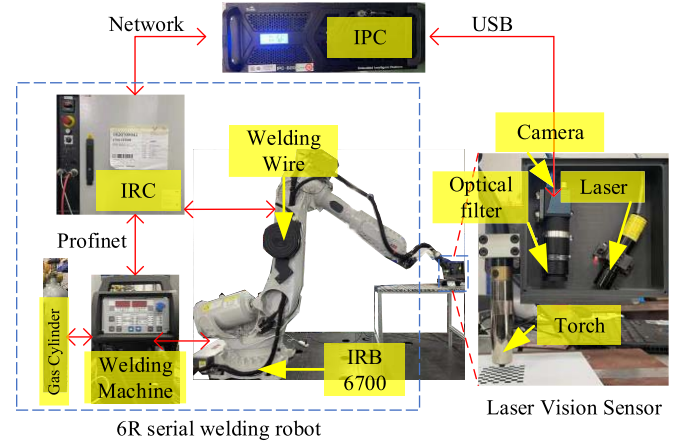


Fig. 2. Experimental equipment of the LVRWS.

equation of the robot end-effector is established. The optical axis constraint endows the instantaneous spatial relationship between “hand” and “eye.” In addition, the angle and distance constraints between the camera and the calibration board are also given.

- 3) To collect diverse and efficient calibration data, we propose a path planning method based on motion constraints and motion selection strategies. In the data acquisition process, the robot’s pose changed greatly, further improving the diversity of the calibration data.

The rest of this article is organized as follows. In Section II, we describe LVRWS and its measurement model. Section III proposes a calibration method based on automatic path planning. In Section IV, the calibration experiments and the weld measurement experiments are conducted to verify the performance of our method. A conclusion drawn from this article is discussed in Section V.

II. SYSTEM DESCRIPTION AND MEASUREMENT MODEL

The LVRWS consists of three subsystems: a six-DoF welding robot system, an industrial personal computer (IPC), and a laser vision sensor, as illustrated in Fig. 2. The laser vision sensor is rigidly mounted on a welding torch forming a hand-eye system. The laser vision sensor sends image data to the IPC at a frequency of 30 Hz. In addition, the IPC controls the robotic trajectory by connecting the externally guided motion (EGM) interface of the robot through network communication. Based on EGM, the IPC can fetch robot data every 4 ms and start executing the movement within 20 ms [30].

A. Coordinate System

In this article, the coordinate system consists of base coordinate system (BCS) F_B , camera coordinate system (CCS) F_C , torch coordinate system (TCS) F_E , calibration board coordinate system (CBCS) F_D as shown in Fig. 3. ${}^A_B T$ in special Euclidean group $SE(3)$ is defined as the homogeneous transformation matrix, representing poses of coordinate system $\{B\}$ at coordinate system $\{A\}$, with the form of

$${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A_B P_B \\ 0 & 1 \end{bmatrix} \quad (1)$$

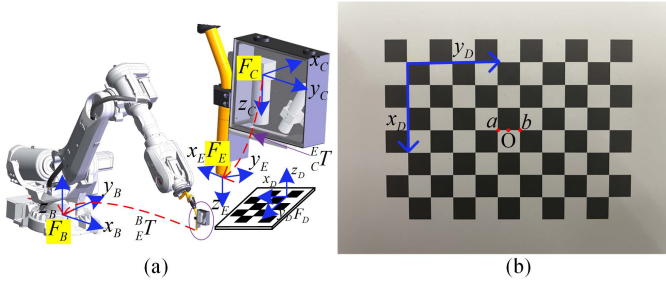


Fig. 3. Coordinate system definition. (a) Coordinate system includes BCS F_B , TCS F_E , and CCS F_C . (b) Coordinate system F_D of the calibration board.

where ${}^A_B R \in SO(3)$, ${}^A P_B \in R^3$ are rotation and translation from the frame B to the frame A .

The pose of the welding torch in the BCS is ${}^B_E T$, which can be controlled or obtained through EGM. The hand-eye transformation is ${}^E_C T$.

B. Measurement Model

The conversion relationship between pixel coordinate $f^t = [u^t, v^t, 1]^T$ (in homogeneous form) in the image pixel coordinate system (IPCS) and 3-D coordinate ${}^B P_f^t = [{}^B P_x^t, {}^B P_y^t, {}^B P_z^t, 1]^T$ (in homogeneous form) in the BCS of laser stripe points at time t is

$${}^B P_f^t = {}^B_E T^t \cdot {}^E_C T \cdot G(f^t) \quad (2)$$

where $G(f^t)$ denotes the transformation that maps a laser stripe point in the IPCS to 3-D coordinate ${}^C P_f^t = [{}^C P_x^t, {}^C P_y^t, {}^C P_z^t, 1]^T$ in the CCS. The hand-eye transformation ${}^E_C T$ and mapping function $G(f^t)$ both need to be calibrated for vision robotic welding system.

The laser plane is parameterized by $ax + by + 1 = c$ in the CCS. The plane parameters are a, b, c , and camera intrinsics K contains f_x, f_y, u_0, v_0 . Camera intrinsics will stay extremely stable in long-term situation, which means that it doesn't need to be calibrated repeatedly. The 3-D coordinate of a laser stripe point f^t in the CCS can be computed by [31]

$${}^C P_f^t = \begin{bmatrix} \frac{u^t - u_0}{f_x} & 0 \\ \frac{v^t - v_0}{f_y} & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c f_x f_y \\ f_x f_y + a f_y (u_0 - u^t) + b f_x (v_0 - v^t) \\ 1 \end{bmatrix}. \quad (3)$$

III. CALIBRATION THEORETICAL METHOD

The calibration framework consists of three stages as shown in Fig. 1. In the first stage, the robot is adjusted to align the camera with the center of the calibration board. In the second stage, an automatic hand-eye calibration procedure is adopted based on simultaneous data collection and optimization. Finally, the linear step motion is used to calibrate the laser light plane automatically.

Fig. 3(b) shows a common flat calibration board, and the midpoint of the corner a and corner b approximately equals the actual center of the checkerboard. The position can be obtained

through exploration motion or manual teaching. Before calibration, the calibration board is placed on the workbench as shown in Fig. 2.

A. Initial Position Adjustment

The first step is aligning the camera optical axis with the calibration board center by adjusting the welding torch position at a fixed attitude.

The camera projection equation of the target in the BCS is

$$\begin{aligned} s \cdot f^t &= K ({}^C_E R ({}^E_B R^t \cdot {}^B P_O + {}^E P_B^t) + {}^C P_E) \\ &= K ({}^C_E R {}^E_B R^t \cdot {}^B P_O + {}^C P_E) + K {}^C_E R {}^E P_B^t \\ &= \text{const} - K {}^C_B R^t \cdot {}^B P_E^t \end{aligned} \quad (4)$$

where s represents the depth of point O in the CCS.

$K ({}^C_E R {}^E_B R^t \cdot {}^B P_O + {}^C P_E)$ and $-K {}^C_B R^t$ are constant matrix when robot's attitude remains unchanged, and ${}^B P_O$ is the position of the calibration board's center in the BCS. Furthermore, if the welding torch only moves inside the xoy plane meaning $\Delta {}^B_E P_z = 0$, the depth s is almost unchanged in that the xoy plane of the CCS is nearly parallel to the calibration board. Thus, according to (4), the differential relationship between the image pixel and the torch translation is

$$\begin{bmatrix} \Delta u \\ \Delta v \\ 0 \end{bmatrix} \approx \frac{1}{s} K {}^C_B R^t \begin{bmatrix} \Delta {}^B_E P_x \\ \Delta {}^B_E P_y \\ 0 \end{bmatrix} = \begin{bmatrix} J_{2 \times 2} & 0_{2 \times 1} \\ M_{1 \times 2} & 0 \end{bmatrix} \begin{bmatrix} \Delta {}^B_E P_x \\ \Delta {}^B_E P_y \\ 0 \end{bmatrix} \quad (5)$$

where J is a constant matrix obtained through the robot's-plane exploratory motion.

We move a certain distance in the x -axis and y -axis directions of the BCS, respectively. At the same time, the pixel coordinates of point O are recorded. Then, two equations are established to solve J .

To adjust the feature point O to the image center C , the robot torch's translation $\Delta {}^B_E P_{xy}$ is given by

$$\begin{bmatrix} \Delta {}^B_E P_x \\ \Delta {}^B_E P_y \end{bmatrix} = J^{-1} \begin{bmatrix} {}^O_C u \\ {}^O_C v \end{bmatrix} \quad (6)$$

where ${}^O_C u$ and ${}^O_C v$ represent the pixel offset from the feature O to image center C in the IPCS.

B. Hand-Eye Automatic Calibration

In this section, the proposed automatic calibration method will be discussed in detail, which mainly consists of three parts: end-effector's motion constraints, path planning, and online nonlinear optimization. The method of automatic path planning and online optimization is conducive to improving calibration efficiency and accuracy.

1) *Motion Constraints of Robot*: As shown in Fig. 4, to ensure that the optical axis of the camera coincides with the center of the calibration board, an equation is derived to compute the torch's translation based on the torch's arbitrary attitude.

To obtain more effective feature points, ensure the center point O of the calibration board is as close to the optical axis of the camera as possible. Noting that the pixel coordinate

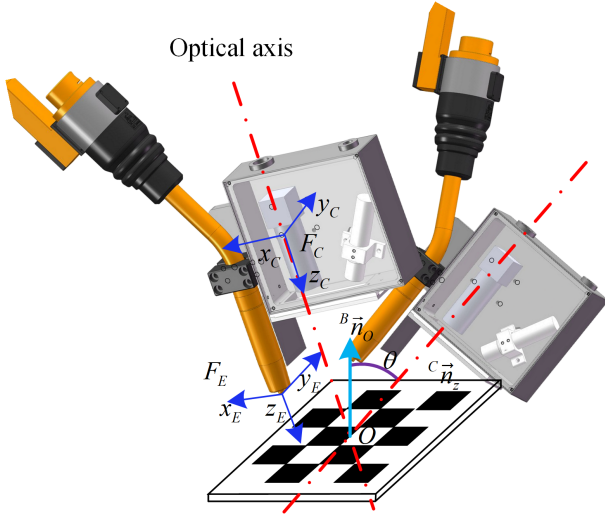


Fig. 4. Hand-eye system on camera's optical axis constraint and rotation constraint.

$p_O = [u_O, v_O, 1]^T$ in the IPCS is approximately equal to the camera principal point $p_0 = [u_0, v_0, 1]^T$

$$p_O = \frac{K}{s} {}^C P_O \approx p_0. \quad (7)$$

Integrating (7) with the transformation relationship of a hand-eye system, we obtain

$$s^i \cdot p_O = K \cdot {}^C E R \cdot {}^E B R_i \cdot ({}^B P_O - {}^B P_E^i) \quad (8)$$

$${}^E B R_i \cdot ({}^B P_E^i - {}^B P_E^i) = {}^E P_C. \quad (9)$$

Combining (8) and (9), we obtain

$${}^E C R \cdot \{s^i \cdot K^{-1} \cdot p_O\} + {}^E P_C = {}^E B R_i \cdot ({}^B P_O - {}^B P_E^i). \quad (10)$$

Assuming s^i and p_O are constant, ${}^E C R \cdot \{s^i \cdot K^{-1} \cdot p_O\} + {}^E P_C$ will become constant vector P_{const} . In addition, under the condition of (7), images that cover all feature points can be collected. Therefore, when the central point of the calibration board is aligned with the optical axis of the camera, the motion constraint equation is as follows:

$$P_{\text{const}} = {}^E B R_i \cdot ({}^B P_O - {}^B P_E^i). \quad (11)$$

Thus, the position of the torch at arbitrary attitude ${}^E B R_i$ based on optical axis alignment constraint is given by

$${}^B P_E^i = {}^B P_O - {}^E B R_i \cdot P_{\text{const}}. \quad (12)$$

The robot pose information is recorded to compute the constant $P_{\text{const}} = {}^E B R^1 \cdot ({}^B P_O - {}^B P_E^1)$ after initial position adjustment. Then, the robot's end-effector motion equation is constructed to obtain the position of the torch on the basis of arbitrary attitude.

To prevent collision, the robot's end-effector must be located at a certain distance d_r above the calibration board. Thus, the distance constraint is

$$({}^B P_E - {}^B P_O) \cdot {}^B \vec{n}_O > d_r \quad (13)$$

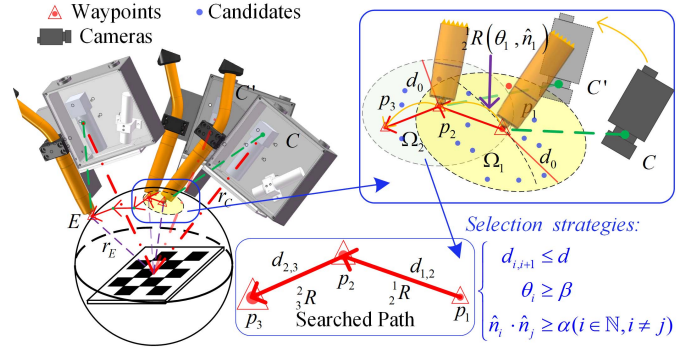


Fig. 5. Path planning based on motion constraints and motion selection strategies.

where ${}^B \vec{n}_O$ is the unit normal vector of the calibration board plane which can be online updated using the principal component analysis (PCA) on all the calibration board's feature points in the BCS. The initial value of ${}^B \vec{n}_O$ is $(0, 0, 1)^T$.

Besides, since the working range of the vision sensor is narrow, it is necessary to restrict the angle θ between the calibration board and the camera to θ_m . The rotation angle θ is computed by

$$\theta = \arccos({}^B \vec{n}_O \cdot \{-{}^E B R {}^E C R {}^C \vec{n}_z\}) \leq \theta_m \quad (14)$$

where ${}^C \vec{n}_z = (0, 0, 1)^T$ represents the CCS's z -axis, θ_m is the maximum angle.

2) *Path Planning*: According to Rodrigues's formula, a rotation matrix R can be described by a rotation axis n and a rotation angle θ . Moreover, ${}^{E_i} E_j R$ and ${}^{C_i} C_j R$ have the same rotation angle, and thus ${}^{E_i} E_{i+1} R$ and ${}^{C_i} C_{i+1} R$ can be expressed as $\text{Rot}(n_{E,i}, \theta_i)$ and $\text{Rot}(n_{C,i}, \theta_i)$, respectively. It is crucial to generate motions sequence for hand-eye iterative optimization.

Since ${}^E P_O$ and ${}^C P_O$ are constant, the camera and torch move on the sphere of radius $\|{}^C P_O\|$ and $\|{}^E P_O\|$ as shown in Fig. 5.

On the basis of error analysis, Zhang et al. [25] summarized the factors affecting the hand-eye calibration error, such as the distance of adjacent "hands," the rotation angle and rotation axis of adjacent "hands." According to the relationship between these factors and errors, the following quantization rules are summarized.

Rule 1: Try to make $\angle(n_{E,i}, n_{E,j})$ no less than α_0 .

Rule 2: Try to make θ_i no less than β_0 .

Rule 3: Try to make $\|{}^{E_i} P_{E_{i+1}}\|$ no bigger than d_0 .

The "rule 2" can be expressed as follows:

$$\theta_i = \arccos\left(\frac{\text{tr}({}^E B R_i {}^E B R_{i+1}) - 1}{2}\right) \geq \beta_0. \quad (15)$$

The "rule 3" can be expressed as follows:

$$d_{i,i+1} = \|{}^B P_{E_{i+1}} - {}^B P_{E_i}\| \leq d_0. \quad (16)$$

Moreover, arbitrary two relative rotations of the end-effector are not coaxial, which dictates "rule 1" as follows:

$$\alpha_{i,j} = \arccos(n_{E,i} \cdot n_{E,j}) = n_{E,i} \cdot n_{E,j} \geq \alpha_0 (i \neq j). \quad (17)$$

Randomly rotate the welding torch around the x -axis, y -axis, and z -axis of the BCS within $\pm 6^\circ$, to generate a point cloud based on motion constraint (11). Then, select a set of points based on the above strategies and parameters α_0, β_0, d_0 to generate a path. As shown in Fig. 5, the waypoint p_1 is the torch initial pose, and the waypoint p_2 is selected based on “rule 1” and “rule 2.” Then, the waypoint $p_{i+2} (i > 0)$ is selected based on all rules. In addition, all waypoints need to be checked using (13) and (14). Noting that the constraint (13) and (14) are refined with each cycle parameter update.

3) *Online Nonlinear Optimization*: We minimize the sum of the Mahalanobis norm of all measurement residuals to obtain a maximum posterior estimation

$$\min_{\chi} \left\{ \sum_{(l,j) \in \mathbb{C}} \left\| r_{\mathbb{C}}(\hat{z}_l^{C_j}, \chi) \right\|_{P_l^{C_j}}^2 \right\}. \quad (18)$$

We use error-state representation [21] to linearize the nonlinear system and solve it with the Levenberg–Marquardt algorithm. In addition, we apply the Mahalanobis norm to achieve robust outlier rejection. The error-state vector becomes

$$\begin{aligned} \delta\chi &= [\delta_C^E T, \delta\lambda_1, \dots, \delta\lambda_m] \\ {}^E_C T &= [{}^E_C R, {}^E P_C] \end{aligned} \quad (19)$$

where m is the total number of features of images, and λ_l is the inverse depth of the l th feature from its first observation. 3-D features are parameterized by their inverse depth λ when first observed in the CCS.

In each Levenberg–Marquardt iteration, (18) is linearized at current state estimation $\hat{\chi}$ to the error state $\delta\chi$

$$\min_{\delta\chi} \left\{ \sum_{(l,j) \in \mathbb{C}} \rho \left(\left\| r_{\mathbb{C}}(\hat{z}_l^{C_j}, \chi) + H_l^{C_j} \delta\chi \right\|_{P_l^{C_j}}^2 \right) \right\} \quad (20)$$

where $H_l^{C_j}$ are Jacobian of visual measurements, and $\rho(s)$ is the Huber norm [32]. Considering that the l th feature is observed in the i th, j th images, the residual [32] is define as

$$\begin{cases} r_{\mathbb{C}}(\hat{z}_l^{C_j}, \chi) = [b_1 \quad b_2]^T \cdot \left(\hat{P}_l^{C_j} - \frac{P_l^{C_j}}{\|P_l^{C_j}\|} \right) \\ \hat{P}_l^{C_j} = \pi_C^{-1} \left(\begin{bmatrix} \hat{u}_l^{C_j} \\ \hat{v}_l^{C_j} \end{bmatrix} \right) \\ P_l^{C_j} = {}^C E R \left({}^E E_j R \left({}^B E_i R \left({}^E_C R \frac{1}{\lambda_l} \pi_C^{-1} \left(\begin{bmatrix} \hat{u}_l^{C_i} \\ \hat{v}_l^{C_i} \end{bmatrix} \right) + {}^E P_C \right) + {}^B P_{E_i} \right) - {}^B P_{E_j} \right) - {}^E P_C \end{cases} \quad (21)$$

where $[\hat{u}_l^{C_i} \quad \hat{v}_l^{C_i}]^T$ is the observation of l th feature that happens in the i th image. $[\hat{u}_l^{C_j} \quad \hat{v}_l^{C_j}]^T$ is the observation of l th feature that happens in the j th image. π_C^{-1} is the inverse-projection function which converts a feature's pixel coordinate into a unit vector using camera intrinsic parameters. b_1, b_2 are two arbitrarily selected orthogonal bases that span the tangent plane of $\hat{P}_l^{C_j}$.

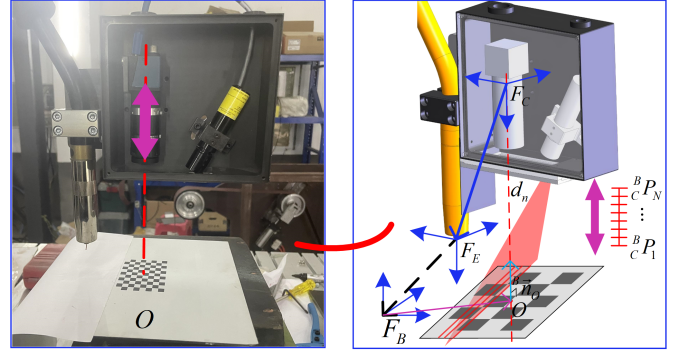


Fig. 6. Line laser plane calibration approach.

Once IPC receives one frame of image, the feature manager thread tries to detect corners of the new image. Then the optimizer thread will update the factor graph by creating a new edge. Finally, the optimizer thread estimates state via the optimization framework Georgia Tech Smoothing and Mapping (GTSAM) [33].

C. Automatic Laser Plane Calibration

The schematic of line laser plane calibration is shown in Fig. 6.

First, we control the position of the robot's end-effector as ${}^B E P_n (n = 0, \dots, N)$ in turn where the camera's z -axis passes through point O , and its attitude is unchanged. Thus, N images of the calibration board and their corresponding laser stripe images are collected. Pixel coordinates (u, v) of points on the laser central stripe can be extracted by the gray centroid method [34]. Then, the 3-D coordinates of laser stripe points can be calculated by method in [9]. In addition, the least square method is used to find the best-fitting plane. The end-effector's target position is defined as follows:

$$\begin{cases} d_n = d_L + d_s \cdot n \\ {}^B E P_n = {}^B P_O + {}^B \vec{n}_O \cdot d_n - {}^B E R_1 {}^E P_C, \quad n \in [0, N] \end{cases} \quad (22)$$

where d_s is the value of moving step, and ${}^B \vec{n}_O$ represents the normal unit vector of the calibration board plane. The distance d_n between the camera origin and the plane meets the constraint $d_L \leq d_n \leq d_H$. $N = (d_H - d_L)/d_s$.

D. Calibration Procedure

The steps of the proposed method are illustrated in Fig. 7. The details of each step are described in the corresponding parts. At the beginning, we open the robotic welding system, EGM, and laser vision sensor. Then, a man-machine interface is used to control the robot welding torch and record the coordinate of the center point O of the calibration board. Automatic calibration for LVRWS is completed in the following steps. First, the plane exploration motion is performed according to Section III-A. After that, the position of the welding torch is adjusted to align the center of the calibration board with the center of the image. Second, automatic hand-eye calibration is performed according to Section III-B

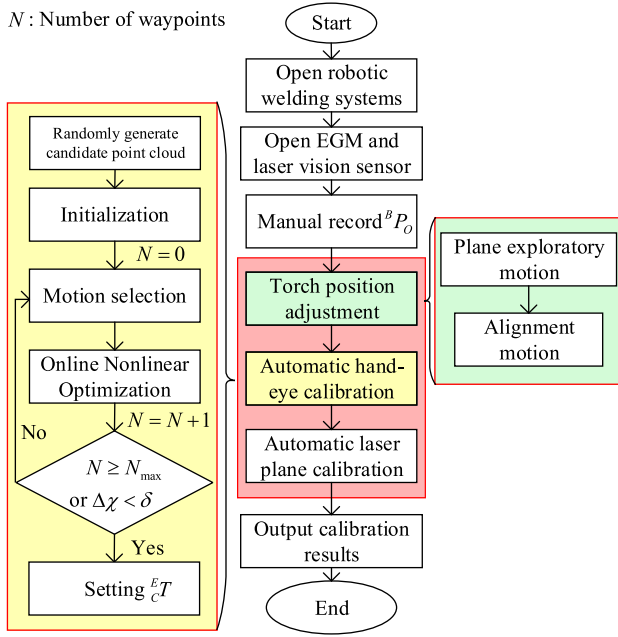


Fig. 7. Calibration procedure.

to obtain the hand-eye matrix. A candidate point cloud is generated by randomly rotating the welding torch. Then, the new waypoint is selected based on the motion selection rules. The robot moves to the waypoint and IPC collects calibration data. Then the nonlinear optimization is performed. If the end condition is met, the loop is closed and the parameters are set; otherwise, the next loop is entered. Third, automatic laser plane calibration is carried out according to Section III-C. Finally, the program outputs the calibration results.

IV. EXPERIMENT AND RESULTS

A. Experiment Configuration

To evaluate the accuracy and efficiency of the proposed method, an LVRWS is set up for experiments.

In the experiment, a six-DoF robot (ABB, IRB6700) is used, which has a repeat positioning accuracy of ± 0.05 mm. The industrial camera in the laser vision sensor is a CMOS camera (DAHENG IMAGING, MER2-503-23GM/C-P, China) with a resolution 2448×2000 pixel. The focal length of the camera lens is approximately 12.42 mm. The laser projector has a wavelength of 660 nm. In addition, the calibration board is a commonly used alumina planar checkerboard with eight rows and 11 columns of grids, and with a side length 6 mm.

As shown in Fig. 8, to evaluate the performance of our calibration algorithm on actual welding workpieces, we used three types of welding workpieces with the curve seam [see Fig. 8(a) and (b)] or line seam [see Fig. 8(c)], and the paths of these weld seams are generated by specific function. First, the positions of key points on the workpiece were obtained through robot teaching. The actual paths were determined by fitting these positions. Then, we taught the robot to scan the welds under various robot poses, while the IPC's program recorded the positions of the feature points.

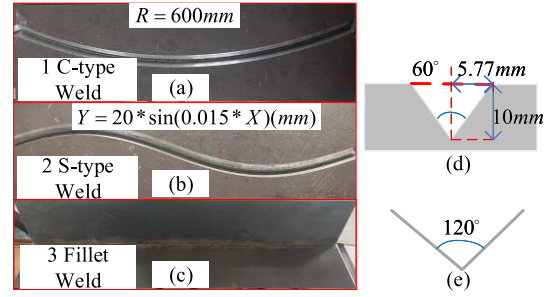


Fig. 8. Three types of welding workpieces. (a) Workpiece 1 is butt weld seam with C-type seam. (b) Workpiece 2 is butt weld seam with S-type seam. (c) Workpiece 3 is a line fillet weld seam. (d) and (e) Cross section of the weld in the tangential direction, and workpiece 1 and 2 have same cross section (d).

To evaluate the static precision of the system, we use the mean absolute error (MAE) for positions. Besides, we use root mean square error (RMSE) of positions to evaluate the stability/robustness of the system.

B. Calibration Experiments and Error Analysis

1) *Torch Initial Position Adjustment*: As shown in Fig. 9(a) and (b), robot explored ($\Delta x = 13.00$ mm, $\Delta y = 0$ mm, and $\Delta z = 0$ mm) in the BCS and calculated pixel increment ($\Delta u = -4$, $\Delta v = 376$) of checkerboard's center O . After that, the robot backed to the initial position. Then, from Fig. 9(a)–(c), robot completed another exploration movement ($\Delta x = 0$ mm, $\Delta y = 12.00$ mm, and $\Delta z = 0$ mm). The equations are formed by substituting the robot's motion and pixel increment into (5). The calculated matrix J is

$$J = \begin{bmatrix} -0.3077 & 28.25 \\ 28.92 & -0.1667 \end{bmatrix}. \quad (23)$$

After determining the matrix J and pixel offset ${}^O_C f = f_C - f_O = (58, 142)$, the robot motion increment $\Delta {}^B_E \hat{P}$ can be calculated using (6) and the center of the calibration board will be initially aligned with the center of the image. Then, $\Delta {}^B_E \hat{P} = J^{-1} {}^O_C f = (5.08, 2.00)$ mm.

Note that the robot's movement is controlled through ABB's EGM interface. To reduce vibration, the target positions of the welding torch need to be changed slowly.

2) *Automatic Hand-Eye Calibration*: After the first step, record the current attitude ${}^B_E R_1$ and position ${}^B_E P_1$, and substitute it into (12) to calculate $P_{\text{const}} = (-63.08, -4.79, 18.97)$. The randomly generated attitude ${}^B_E R_i$ are substituted into (12) to obtain the corresponding positions and form the point cloud of the candidate pose.

According to the motion selection strategies, the continuous points set was selected for hand-eye calibration based on the randomly generated candidate points under motion constraints as shown in Fig. 10. As shown in Fig. 11, blue points represent candidate path points and the red line is the generated path.

3) *Automatic Laser Plane Calibration*: By adjusting the exposure time of the camera, the laser fringe image and the calibration board image are collected at the same position, respectively. As shown in Fig. 12, the exposure time of the calibration board image is set to 50 ms, and the exposure

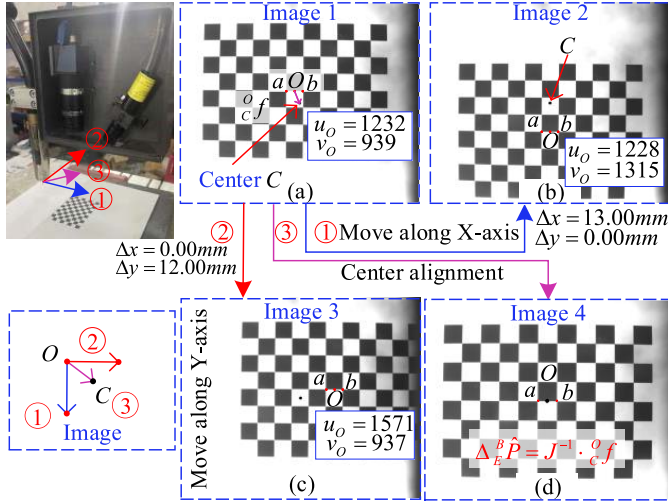


Fig. 9. Illustration of torch initial position adjustment. (a) Image taken at the initial position. (b) Image after x-direction exploration. (c) Image after y-direction exploration. (d) Image after initial alignment.

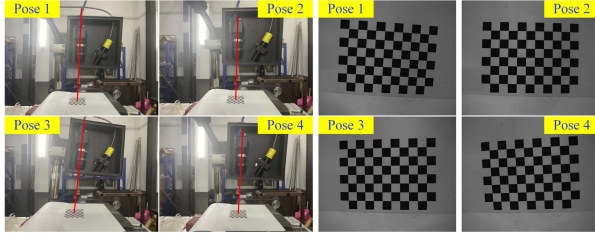


Fig. 10. Pictures of laser vision sensor at multiple poses based on optical axis constraint.

The trajectory of robot end-effector and candidate path points

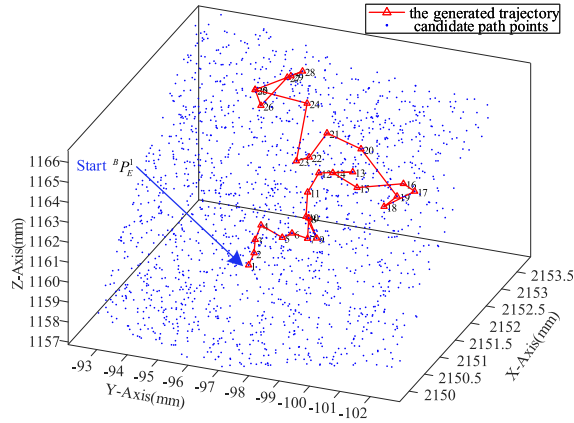


Fig. 11. Generated path of the end-effector from candidate path points ($d = 0.5$ mm, $\alpha = 1^\circ$, $\beta = 3^\circ$).

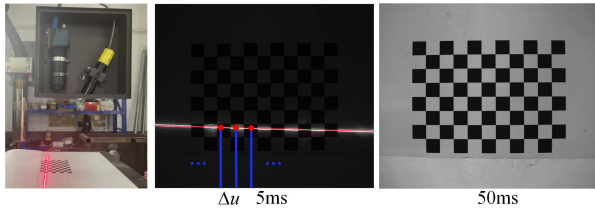


Fig. 12. Laser stripe image and calibration board image are collected at different positions, respectively.

time of the laser fringe image is set to 5 ms. The linear stepping motion is set to $d_L = 120.00$ mm, $d_H = 134.00$ mm, $d_s = 2.00$ mm.

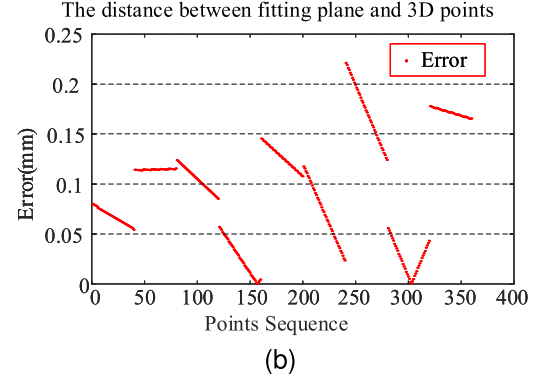
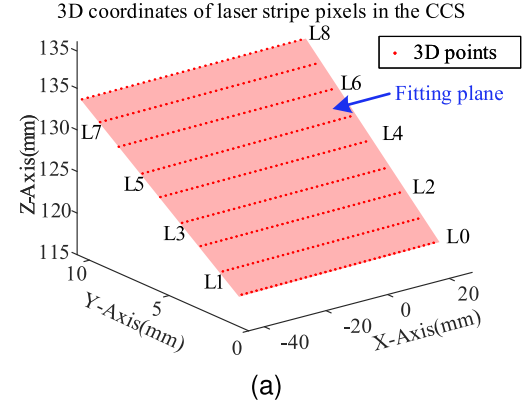


Fig. 13. Line laser light plane calibration. (a) Laser plane 3-D points in the CCS. (b) Laser plane fitting residual error.

TABLE I
MEASUREMENT ATTITUDES (XYZ EULER ANGLES)

NAME	X (degree)	Y (degree)	Z (degree)
O1	0	174	0
O2	-6	0	9
O3	6	-174	-9
O4	-6	174	9
O5	6	-174	9

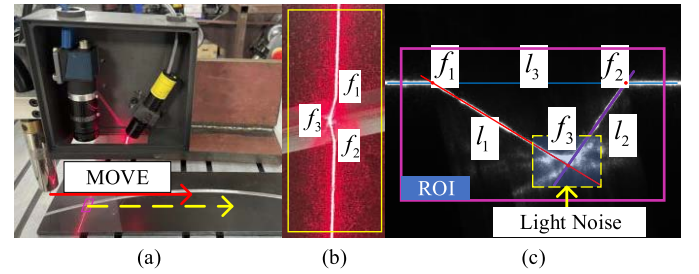


Fig. 14. Measurement of the curve welding workpieces. (a) Laser line vision sensor scanning the weld. (b) Laser stripe projected on the weld. (c) ROI of laser stripe image and the key feature point's analysis.

Fig. 13(a) shows the 3-D point cloud. The least square fitting method is used to fit the plane equation in CCS. The plane equation is as follows:

$$-0.00964276x + 1.81429y + z = 114.353. \quad (24)$$

As shown in Fig. 13(b), the errors are the Euclidean distance from the 3-D points to the light plane. The MAE and RMSE

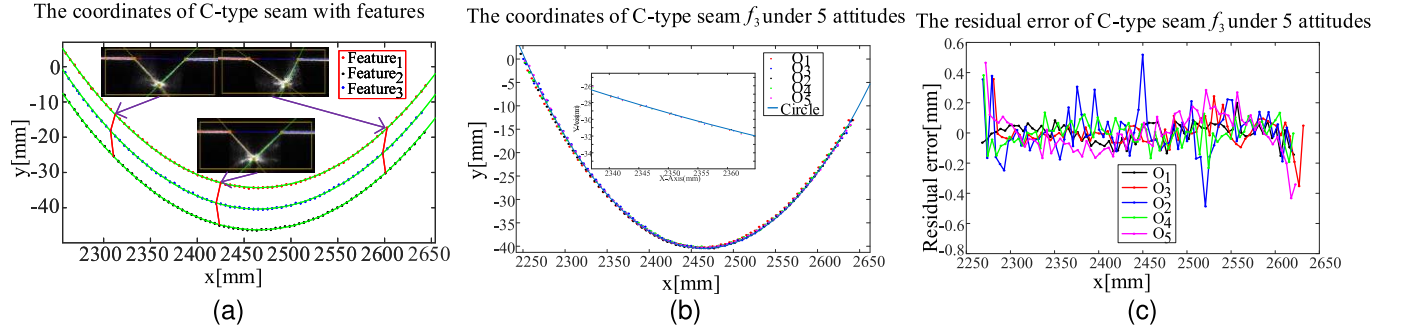


Fig. 15. Experimental results of test set 1. Note: Coordinates of the laser vision sensor measurements on the xoy plane in the BCS, and the ground truth is depicted in green solid line (workpiece 1, C-type seam, five types of attitudes). (a) Coordinates of laser stripe image's feature points, and detection samples of the laser stripe image sequences. (b) Coordinates of the laser stripe feature f_3 under five attitudes, and the circle fitting path of teaching points (green solid curve). (c) Residual error. The proposed method is indeed accurate for dynamic welding.

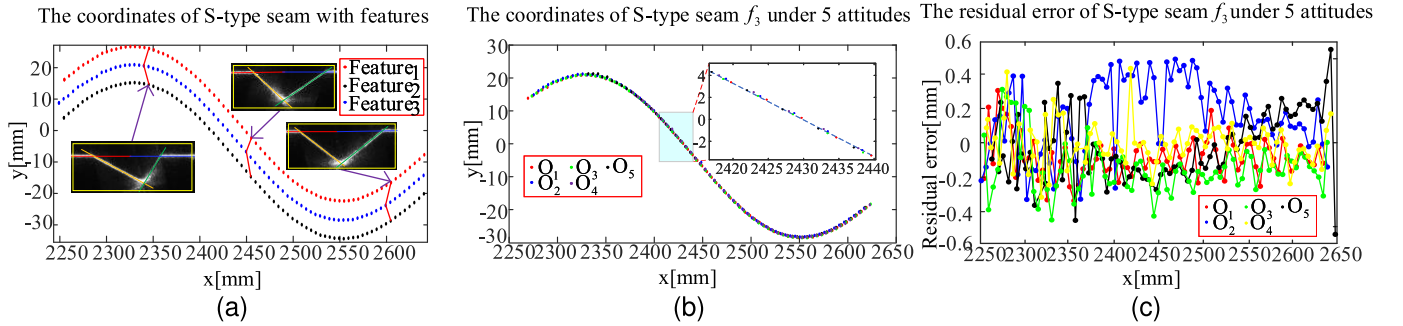


Fig. 16. Experimental results of test set 1. Note: Coordinates of the laser vision sensor measurement on the xoy plane in the BCS (workpiece 2, S-type seam, five types of attitudes). (a) Coordinates of the laser stripe features, and detection samples of the laser stripe image. (b) Coordinates of the laser stripe feature f_3 under five attitudes. (c) Residual error.

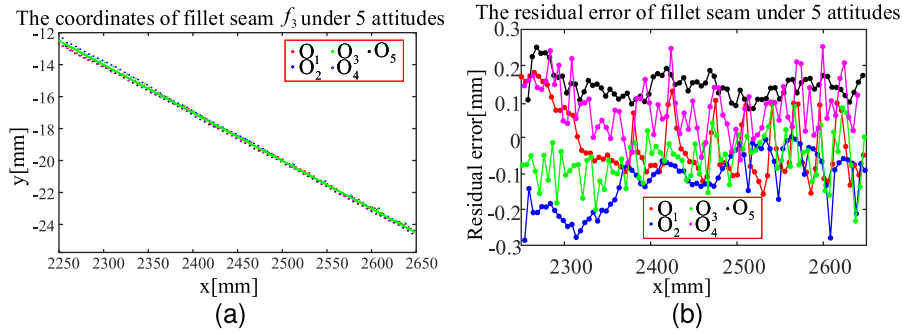


Fig. 17. Experimental results of test set 1. Note: Coordinates of the laser vision sensor measurements on the xoy plane in the BCS (workpiece 3, line seam, five types of attitudes). (a) Coordinates of the laser stripe feature under five attitudes ("TP" represents teaching point, green solid line represents the ground truth of the seam). (b) Residual error.

of the proposed method are 0.0974 and 0.1115 mm, which is superior to [9]'s work. The significant decrease in plane fitting error is attributed to the enlargement of laser stripe resolution, which enhances the precision of laser stripe extraction. In addition, points on the same laser stripe have linear errors. This is because the pose estimation of the calibration board has systematic error.

C. Dynamic Measurement Experiments

The seam measurement experiments were conducted to evaluate the dynamic performance. The robot was driven to

scan along the seam under various attitudes, and the calibrated line laser vision sensor was used to detect the seam of the welding workpieces. Then, the line weld and C-type weld were measured by robot teaching. Besides, as shown in Table I, five types of attitudes O_i ($i = 1, 2, 3, 4, 5$) in the BCS were selected to verify the dynamic performance.

The laser stripe projected on the curve welds contains three feature points f_1, f_2, f_3 as shown in Fig. 14(b). In addition, fillet welds have only one feature point. Furthermore, the region of interest (ROI) of the laser stripe image is illustrated in Fig. 14(c). The detection of feature points mainly draw support from the central line intersection approach [35]. In particular,

TABLE II
MAE AND RMSE WITH THREE WORKPIECES

ERROR	C-type		S-type		Fillet	
	MAE (mm)	RMSE (mm)	MAE (mm)	RMSE (mm)	MAE (mm)	RMSE (mm)
O ₁	0.1285	0.0892	0.0980	0.0974	0.0863	0.0942
O ₂	0.1175	0.1710	0.1584	0.1287	0.1206	0.1427
O ₃	0.1301	0.1641	0.1684	0.1958	0.1426	0.1469
O ₄	0.1627	0.1546	0.1512	0.1767	0.1367	0.1225
O ₅	0.1484	0.1772	0.1304	0.1607	0.1502	0.1395

since feature f_3 is polluted by strong light noise, robust extraction of feature points achieves accurate weld measurement.

As shown in Figs. 15(a) or 16(a), the detected feature points f_1, f_2, f_3 from laser stripe images are converted to coordinates in the BCS.

In Fig. 15(b), the coordinates of the C-type seam teaching points on the xoy plane are fitted with the circular fitting algorithm. The circular equation is used to calculate the residual as shown in Fig. 15(c). For the line seam, the linear fitting of two teaching points is used to solve the line equation. As for S-type weld, due to the pose's uncertainty of the weld seam in the BCS, it is difficult to fit the sine curve. Hence, linear fitting is performed on a small window (2-cm length on x -axis) to approximate the ground truth.

The residual of S-type weld shown in Fig. 16(c) indicates that the absolute value of residuals is below 0.6 mm and the measurement under various attitudes has high repeatability. This is consistent with the C-type weld. Besides, for fillet weld, the absolute value of residuals is within 0.3 mm as shown in Fig. 17(b). And the MAE and RMSE are demonstrated in Table II. As can be seen from Table II, MAE and RMSE of positions are both less than 0.20 mm. The measurement accuracy of straight line weld is similar to that of curve weld, but the maximum absolute error of straight line weld is less than that of curve weld. The most important reason for this is that the straight stripe of curve weld images is much shorter than that of line weld images.

In addition, the measurement accuracy is consistent in different robot attitudes, which implies that the calibrated system has good repeatability. In previous work [11], the weld positioning error of the six-DoF welding robot was controlled within 0.8 mm. In contrast, the maximum absolute error of the proposed method is less than 0.6 mm. Moreover, MAE of our weld positioning error is within 0.2 mm, which is essentially equivalent to prior work [12]. In summary, compared with existing tedious calibration methods, the proposed algorithm can achieve the same calibration accuracy.

In addition, one procedure of calibration can be completed within 2 min, which only depends on one initial manual operation. Compared with previous methods that require a complex on-site operation, the proposed method indeed improves the efficiency of data acquisition or calibration.

The method proposed in this article mainly solves the problem of automatic calibration without teaching when the hand-eye relationship is unknown. The innovation of the algorithm in this article is more reflected in efficiency and

simplicity. Therefore, we directly compared the weld seam measurement accuracy with those in available literature.

To sum up, the accuracy and efficiency of our proposed automatic calibration method meet the requirements of dynamic welding.

V. CONCLUSION

Aiming at the problem that the manual calibration process of LVRWS is complex and inefficient, an automatic calibration method is proposed in this article. First, motion constraint to the optical axis of the camera is derived, and rotation constraint based on the camera's observation field is given. Then, an automatic path planning method is presented, and online optimization based on a maximum posterior estimator is adopted to estimate the parameters. Finally, the accuracy experiments with multitype weld are performed. The MAE and RMSE of position for curved and fillet welds are less than 0.20 mm under complex welding torch's attitude. The results show that the proposed methods have a distinct improvement in feasibility, efficiency, and adaptability.

In the future, to improve the performance of LVRWS, systematic errors due to vibration should be considered.

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